ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)			**Assignment 4**
Branch	CSIT	Programme	**B.Tech**
Course Name	Full-Stack Web Development with MERN	Semester	6^{th}
Course Code	CSE 3839	Academic Year	2025-2026
Lab Assignment: 4		**Topic: JWT fundamentals and backend authentication**	

Learning Level (LL)			
	L1: Remembering	**L3:** Applying	**L5**: Evaluating
	L2: Understanding	**L4**: Analysing	**L6**: Creating

Q's	Questions	COs	LL
1.	A company is developing a Product Store frontend application using React.js. The application should contain different frontend forms to manage user authentication and product searching functionality. (a) Create a LoginForm React component containing: Email field, Password field, Login button The form should collect user login credentials and display appropriate validation messages. (b) Develop a RegisterForm React component containing: Username field Email field Password field Confirm Password field Register button The form should validate password matching before submission. (c) Create a ProductSearchForm React component containing: Product name search field Category dropdown Search button Configure TanStack Query to fetch and display products based on the search criteria entered by the user.	**CO1,CO2, CO3,CO4**	**L3 L4**
2.	A startup company is developing a Product Store application using the MERN	**CO1, CO3,CO5**	**L3 L4**

	stack without using any database. The application must allow users to register and log in securely using JSON Web Token authentication. Registered users should receive a JWT token after successful login and use it to access a protected products route. (a) Develop a backend registration and login API using Express.js that stores user details temporarily in an array and generates a JWT token for valid login credentials. (b) Design a React frontend containing Register and Login forms that communicate with the backend using the Fetch API and store the JWT token in localStorage after successful login. (c) Implement a protected /products API route that verifies the JWT token before displaying product information, and test the APIs using Postman.		

1. Assignment carries a weightage of 20 marks out of 100
2. CO1,CO2,CO3,CO4 and CO5 outcomes are covered.

Course Outcomes	CO1	Understand and configure a modern full-stack development environment using Node.js, MongoDB, Docker, and supporting tools.
	CO2	Design and implement RESTful backend services using Express, MongoDB (Mongoose), and unit testing frameworks.
	CO3	Develop interactive frontend applications using React and efficiently integrate them with backend APIs using modern data-fetching techniques.
	CO4	Apply DevOps practices to containerize, test, and deploy full-stack applications using Docker and CI/CD pipelines.
	CO5	Implement authentication, performance optimization, Search Engine Optimization techniques, and automated testing for scalable and secure web applications.
	CO6	Build and integrate advanced full-stack architectures using data aggregation, visualization, and GraphQL APIs with modern frontend clients.